A
Project Based Learning
Report on

Liver Disease Prediction

Submitted to

ITM UNIVERSITY
GWALIOR • MP • INDIA

For the partial fulfillment of the requirement for the course

Machine Learning
(CSE0314)

BACHELOR OF TECHNOLOGY
in
AIML
for
Academic Session: 2026–27

Submitted by

R.Mahesh-BETN1AI25204
R.Barath-BETN1AI25205
P.Vamsi-BETN1AI25200

B.Tech-III
Section-F

Under the Supervision of

Mr.Ashish SinghYadav
Assistant Professor
SoCIS,
ITM UNIVERSITY,
GWALIOR (M.P)

ABSTRACT

This project develops and compares machine-learning classifiers for predicting the liver-disease class recorded in the Indian Liver Patient Dataset (ILPD). The notebook uses a CSV file containing 583 patient records, 10 input attributes and one target column. The attributes include age, gender, bilirubin measurements, alkaline phosphotase, aminotransferase measurements, total proteins, albumin and the albumin-to-globulin ratio. Initial inspection identified four missing values in Albumin_and_Globulin_Ratio and 13 duplicate rows. After duplicate removal, 570 records remained. Gender and target labels were encoded, the data was divided into stratified training and test subsets (80:20), missing feature values were imputed using the training-set median, and RandomOverSampler was applied only to the training data to balance the two classes. Three classifiers were trained: Decision Tree, Random Forest and Gaussian Naive Bayes. On the 114-record test set, Random Forest achieved the highest accuracy (0.711) and F1-score (0.807), with precision 0.767 and recall 0.852 for the positive liver-disease class. Decision Tree achieved an F1-score of 0.786, while Gaussian Naive Bayes achieved 0.566. Random Forest was therefore selected as the best-performing model among the three experiments. The project demonstrates a complete introductory classification workflow, but its small dataset, moderate test performance, lack of external validation and absence of systematic hyperparameter tuning limit its use. The output is an academic experiment and is not a medical diagnosis or clinical decision tool.

TABLE OF CONTENTS

Section	Page No.
Abstract	ii
List of Figures	iii
List of Tables	iii
List of Abbreviations	iii–iv
Chapter 1 – Introduction	1
Chapter 2 – Literature Review	3
Chapter 3 – Dataset Description and Data Understanding	5
Chapter 4 – Data Preprocessing and Exploratory Data Analysis	8
Chapter 5 – Proposed Methodology and ML Model Development	13
Chapter 6 – Model Evaluation and Performance Analysis	16
Chapter 7 – Results and Discussion	20
Chapter 8 – Conclusion and Future Scope	22
References / Bibliography	23
Appendix A – Dataset Information	24
Appendix B – Additional Results	24
Appendix C – Project Screenshots	24
Appendix D – GitHub Repository	24
Appendix E – Team Contributions	24

LIST OF FIGURES

Figure 3.1 – Dataset class distribution
Figure 3.2 – Gender distribution
Figure 4.1 – Box plots of numerical features
Figure 4.2 – Age distribution
Figure 4.3 – Correlation heatmap
Figure 4.4 – Training class distribution after oversampling
Figure 5.1 – Decision Tree visualization
Figure 6.1 – Decision Tree confusion matrix
Figure 6.2 – Random Forest confusion matrix
Figure 6.3 – Gaussian Naive Bayes confusion matrix
Figure 6.4 – Model performance comparison

LIST OF TABLES

Table 3.1 – Dataset summary
Table 3.2 – Feature descriptions
Table 3.3 – Initial data-quality assessment
Table 3.4 – Descriptive statistics
Table 4.1 – Preprocessing summary
Table 5.1 – Machine-learning algorithms
Table 5.2 – Model configuration
Table 6.1 – Evaluation metrics
Table 6.2 – Confusion matrices
Table 6.3 – Model performance comparison
Table 7.1 – Sample-record prediction outputs

LIST OF ABBREVIATIONS

Abbreviation	Meaning
ILPD	Indian Liver Patient Dataset
ML	Machine Learning
EDA	Exploratory Data Analysis
CSV	Comma-Separated Values
ALT / SGPT	Alanine Aminotransferase
AST / SGOT	Aspartate Aminotransferase
ALP	Alkaline Phosphotase
A/G Ratio	Albumin-to-Globulin Ratio
TP	Total Proteins
RF	Random Forest
DT	Decision Tree
NB	Naive Bayes
TN / TP	True Negative / True Positive
FP / FN	False Positive / False Negative

ALT / SGPT	Alanine / Alamine Aminotransferase
AST / SGOT	Aspartate Aminotransferase
ALP	Alkaline Phosphotase
A/G Ratio	Albumin-to-Globulin Ratio
TP	Total Proteins
RF	Random Forest
DT	Decision Tree
NB	Naive Bayes
TN / TP	True Negative / True Positive
FP / FN	False Positive / False Negative

CHAPTER 1: INTRODUCTION

1.1 Introduction to the Project Domain

Machine learning is a branch of artificial intelligence that enables computer systems to learn patterns from data and use those patterns to make predictions or classifications. In healthcare-related data analysis, classification methods can be used to study relationships between patient measurements and an existing diagnostic label. This project applies supervised machine learning to the Indian Liver Patient Dataset (ILPD), where each row represents a patient record and the target column indicates one of two classes.

1.2 Background of the Problem

The liver performs several essential functions, including metabolism, protein production and processing of substances in the blood. Clinical measurements such as bilirubin, alkaline phosphotase, aminotransferase levels, total proteins and albumin are included in the selected dataset. The classification task is to learn patterns in these measurements that distinguish records labelled as liver disease from records labelled as no liver disease. The UCI repository describes ILPD as a classification dataset containing 583 instances and 10 features, collected from patients in the north-east region of Andhra Pradesh, India .

1.3 Motivation

The motivation is to understand how a small, real-world medical dataset can be prepared and used to compare different classification algorithms. The project also provides practical experience in data cleaning, missing-value treatment, class-imbalance handling, model training and performance evaluation.

1.4 Problem Statement

Given a patient record containing age, gender and nine biochemical measurements, develop and compare machine-learning classification models that predict the binary target label used in the ILPD file. The objective is to identify which of the implemented models provides the strongest performance on a held-out test set.

1.5 Project Objectives

- Load and understand the ILPD CSV dataset and its attributes.
- Identify missing values, duplicate records and class imbalance.
- Prepare the data through duplicate removal, categorical encoding, median imputation and training-only oversampling.
- Perform exploratory analysis using distributions, box plots and a correlation heatmap.
- Implement Decision Tree, Random Forest and Gaussian Naive Bayes classifiers.

- Evaluate the models using accuracy, precision, recall, F1-score and confusion matrices.
- Compare the results and select the best-performing model based on the observed metrics.

1.6 Scope of the Project

The project scope is limited to an academic binary-classification experiment using the provided ILPD CSV file and the three algorithms implemented in the notebook. It includes exploratory analysis, preprocessing, model training, test-set evaluation and a sample-record demonstration. It does not include clinical validation, deployment in a hospital, diagnosis, treatment advice, prospective testing or a production-grade medical application.

1.7 Organization of the Report

Chapter 1 introduces the problem and objectives. Chapter 2 reviews the dataset and relevant machine-learning approaches. Chapter 3 describes the dataset and its quality. Chapter 4 explains preprocessing and EDA. Chapter 5 presents the workflow and models. Chapter 6 evaluates and compares model performance. Chapter 7 discusses the experimental results. Chapter 8 concludes the work and outlines limitations and future improvements.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction to Existing Research

The ILPD dataset is commonly used as a benchmark for supervised classification research involving liver-disease labels. The dataset contains demographic and biochemical attributes, making it suitable for comparing models that learn decision boundaries from tabular data. The UCI Machine Learning Repository provides the dataset description, feature definitions, task type and citation information .

2.2 Review of Related Research Papers

Straw and Wu (2022) examined potential sex-related bias in supervised machine-learning models for liver-disease prediction using ILPD. Their work highlights that model performance should not be considered only as a single overall score; performance across patient subgroups can also matter in healthcare applications [2]. This is relevant to the present project because gender is one of the input attributes. However, the current notebook does not conduct a sex-stratified fairness analysis, so no subgroup fairness conclusion is claimed.

Published work using ILPD has explored different classification and data-mining approaches. Such studies provide context for using the dataset to examine relationships between biochemical attributes and the class label. The present project takes a comparative learning approach by implementing three commonly taught classifiers in one consistent notebook workflow.

2.3 Existing Approaches and Techniques

- Decision Tree classifiers use feature-based splitting rules to partition records into groups and assign a class at a leaf node.
- Random Forest combines predictions from multiple decision trees. The ensemble can reduce dependence on a single tree and model more varied patterns.
- Gaussian Naive Bayes applies Bayes' theorem and models each numerical feature with a Gaussian likelihood under a conditional-independence assumption.
- Data preparation techniques such as median imputation, categorical encoding and random oversampling are used to make the dataset suitable for model training.

2.4 Limitations of Existing Approaches

A model trained on a small benchmark dataset may not generalize to different hospitals, populations, measurement devices or clinical settings. Overall accuracy can also hide class-specific errors. Class imbalance may cause a classifier to favour the majority class, while oversampling can change the training distribution. For this reason, evaluation should include precision, recall, F1-score and confusion matrices rather than relying on accuracy alone.

2.5 Research Gap / Project Motivation

The project does not claim to introduce a new algorithm. Its practical focus is a reproducible comparison of three different classification approaches after a documented preprocessing workflow. It compares their test-set metrics and examines the trade-offs between identifying positive liver-disease records and avoiding false-positive predictions. A further gap in the current implementation is the absence of subgroup analysis, cross-validation, external validation and systematic hyperparameter optimization; these are identified as future work.

CHAPTER 3: DATASET DESCRIPTION AND DATA UNDERSTANDING

3.1 Dataset Introduction

The project uses the Indian Liver Patient Dataset (ILPD). The UCI repository describes the dataset as a multivariate classification dataset with 583 instances and 10 input features [1]. The executed notebook reads a CSV file named indian_liver_patient.csv and reports an initial shape of 583 rows and 11 columns, consisting of ten input attributes and one target column.

3.2 Dataset Source and Reference Link

The CSV was downloaded from Kaggle . The UCI dataset record is available at: https://www.kaggle.com/datasets/uciml/indian-liver-patient-records.

3.3 Dataset Size and Structure

Parameter	Description
Dataset	Indian Liver Patient Dataset (ILPD)
CSV file	indian_liver_patient.csv
Initial shape in notebook	583 rows × 11 columns
Input features	10
Target column	Dataset
Problem type	Binary classification
Rows after duplicate removal	570
File format	CSV

3.4 Feature / Attribute Description

Attribute	Type	Description
Age	Integer	Age of the patient in years
Gender	Categorical	Male or Female; encoded as Male=1, Female=0
Total_Bilirubin	Continuous	Total bilirubin measurement
Direct_Bilirubin	Continuous	Direct bilirubin measurement
Alkaline_Phosphotase	Integer	Alkaline phosphotase enzyme measurement
Alamine_Aminotransferase	Integer	ALT / SGPT enzyme measurement
Aspartate_Aminotransferase	Integer	AST / SGOT enzyme measurement
Total_Protiens	Continuous	Total protein measurement
Albumin	Continuous	Albumin measurement
Albumin_and_Globulin_Ratio	Continuous	Albumin-to-globulin ratio

3.5 Target Variable Description

The target column is named Dataset. In the original CSV coding used by the notebook, label 1 represents liver disease and label 2 represents no liver disease. The notebook maps label 1 to 1 and label 2 to 0. Thus, in the prepared target, 1 = Liver Disease and 0 = No Liver Disease.

3.6 Data Types and Statistical Summary

The notebook reports five floating-point columns, five integer columns and one object column before encoding. The descriptive statistics below are from the original 583-row dataset before duplicate removal.

Attribute	Mean	Std. Dev.	Minimum	Maximum
Age	44.75	16.19	4	90
Total_Bilirubin	3.30	6.21	0.4	75
Direct_Bilirubin	1.49	2.81	0.1	19.7
Alkaline_Phosphotase	290.58	242.94	63	2110
Alamine_Aminotransferase	80.71	182.62	10	2000
Aspartate_Aminotransferase	109.91	288.92	10	4929
Total_Protiens	6.48	1.09	2.7	9.6
Albumin	3.14	0.80	0.9	5.5
Albumin_and_Globulin_Ratio	0.95	0.32	0.3	2.8

3.7 Initial Data Quality Assessment

Quality check	Notebook observation	Action / status
Missing values	4 missing values, all in Albumin_and_Globulin_Ratio	Median imputation performed after split
Duplicate rows	13 duplicate rows	Removed; 570 records remained
Outliers	Box plots show wide ranges and extreme values in some enzyme measures	Detected visually; no outlier removal or capping recorded
Class imbalance	406 class 1 and 164 class 2 after duplicate removal	Training data balanced using RandomOverSampler
Inconsistent values	No separate inconsistent-value correction step record	Not claimed as corrected

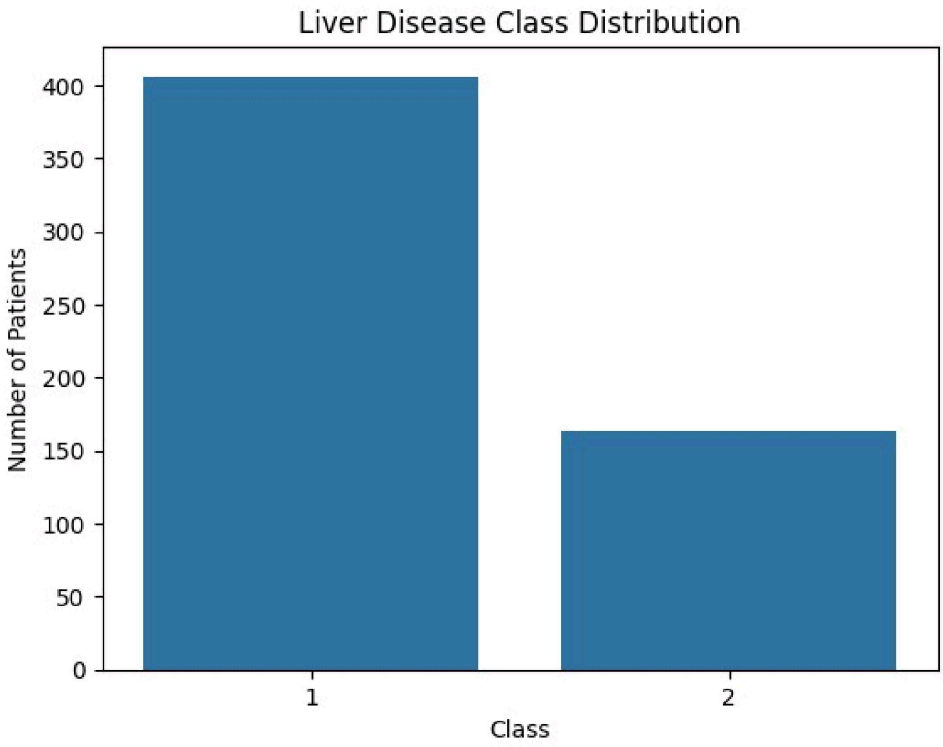


Figure 3.1: Class distribution after duplicate removal.

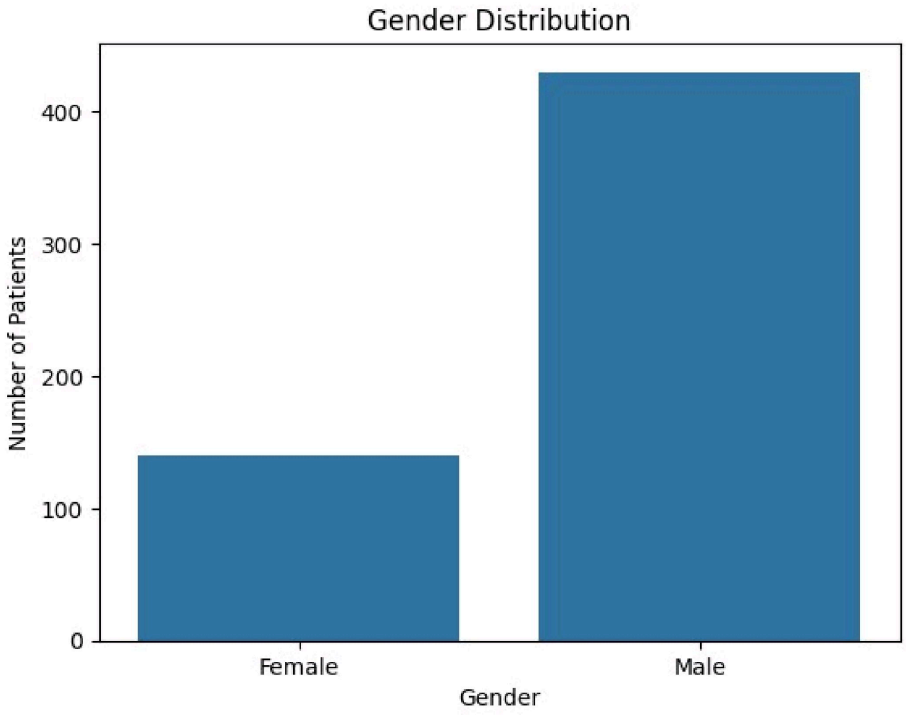


Figure 3.2: Gender distribution after duplicate removal.

The dataset after duplicate removal contains 406 positive liver-disease records and 164 negative records. This imbalance motivates balancing the training subset before model fitting. The notebook reports 430 male and 140 female records after duplicate removal.

CHAPTER 4: DATA PREPROCESSING AND EXPLORATORY DATA ANALYSIS

4.1 Data Cleaning

The notebook loads the CSV using pandas and checks its shape, data types, statistical summary, missing values and duplicate records. Duplicate rows are removed using the drop_duplicates() operation. The dataset changes from 583 × 11 to 570 × 11, and a subsequent check confirms zero remaining duplicates.

4.2 Handling Missing Values

Four missing values are found in Albumin_and_Globulin_Ratio. A SimpleImputer configured with strategy='median' is fitted on the training features and then used to transform both training and test features. This avoids calculating the imputation value from the test set. After imputation, the notebook reports zero missing values in both subsets.

4.3 Duplicate Removal

A total of 13 duplicate rows are identified and removed before the train-test split. The resulting dataset contains 570 unique rows.

4.4 Outlier Detection and Treatment

Box plots are generated for the numerical attributes to inspect their spread and identify potential extreme observations. Some biochemical variables have broad ranges, including high maximum values for aminotransferase measurements. The notebook uses box plots for visual detection but does not apply a documented outlier-removal, winsorization or capping procedure. Therefore, this report records outlier detection, not outlier treatment.

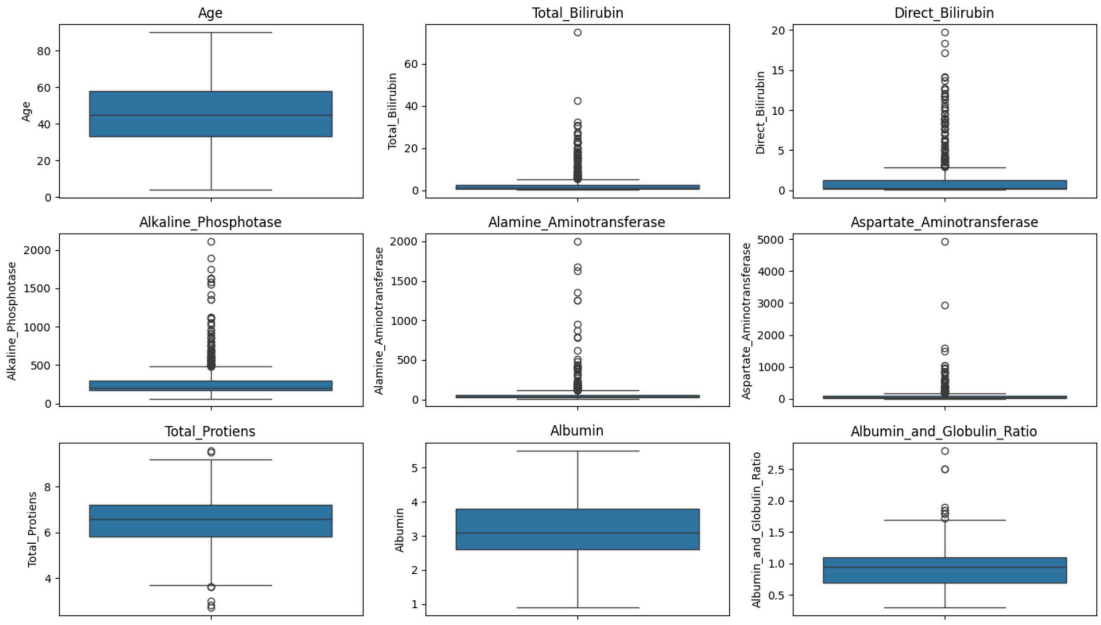


Figure 4.1: Box plots of numerical features used for visual outlier inspection.

4.5 Categorical Data Encoding

The Gender column is converted to numeric values using Male = 1 and Female = 0. The target column is converted using original label 1 → 1 (Liver Disease) and original label 2 → 0 (No Liver Disease).

4.6 Feature Scaling / Normalization

No feature-scaling or normalization operation is present in the executed notebook. The models are trained using the imputed numerical feature values directly. This is recorded as an implementation choice rather than claiming that scaling was performed.

4.7 Feature Selection / Engineering

The target column Dataset is separated from the ten predictor attributes. No separate feature-selection algorithm, dimensionality-reduction method or engineered feature is recorded in the notebook. All ten input attributes are retained for model training.

4.8 Train-Test Split

The data is divided using train_test_split with test_size=0.20, random_state=42 and stratify=y. The resulting training subset contains 456 rows and the test subset contains 114 rows. Stratification preserves the approximate class proportions in both subsets.

Subset	Total rows	Liver disease (1)	No disease (0)
Training before oversampling	456	325	131
Testing	114	81	33
Training after oversampling	650	325	325

4.9 Exploratory Data Analysis (EDA)

EDA is performed using pandas, Matplotlib and Seaborn. The notebook generates a target-class count plot, gender count plot, age histogram with a kernel-density curve, box plots for numerical attributes and a correlation heatmap. These plots are used to understand the class distribution, demographic distribution, feature spread and pairwise numeric correlations.

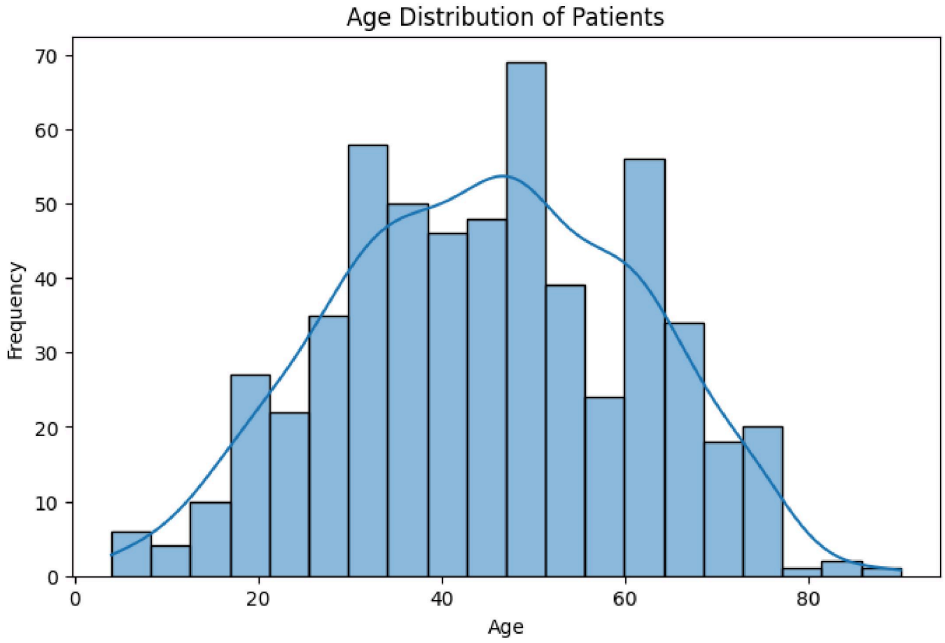


Figure 4.2: Age distribution of patient records.

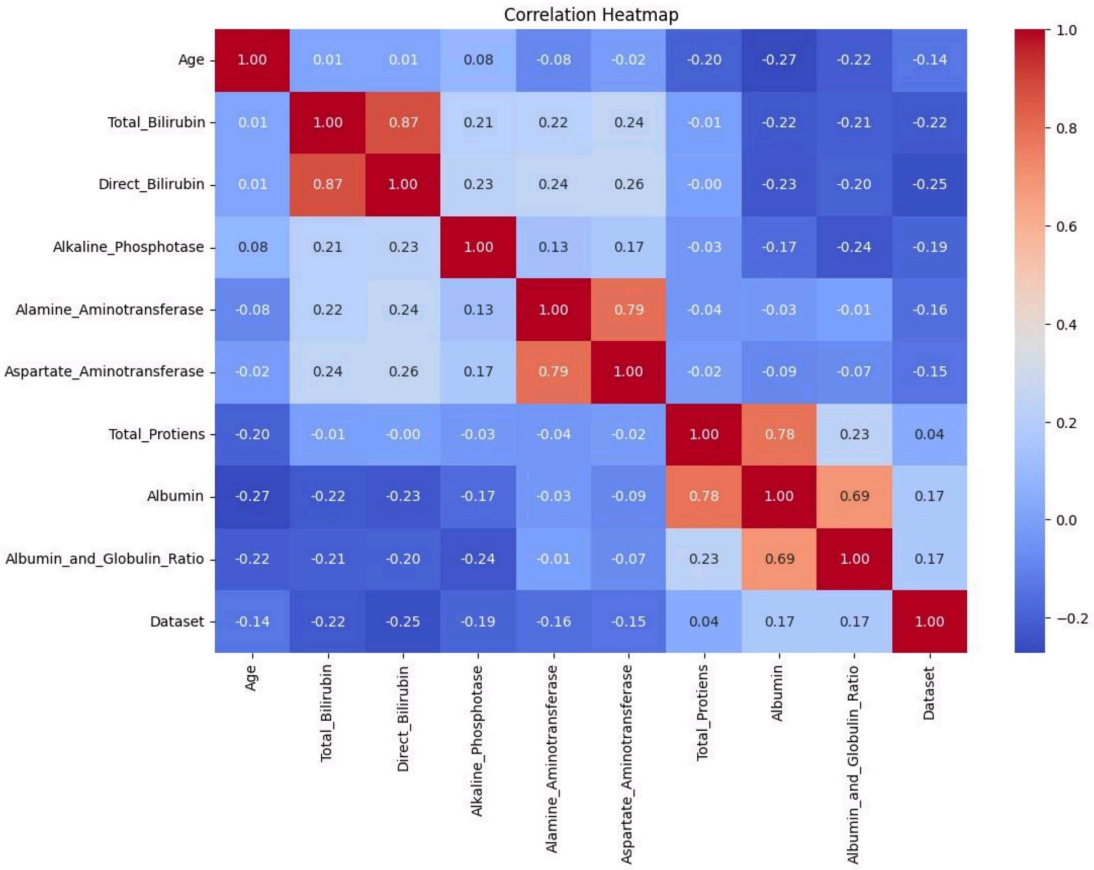


Figure 4.3: Correlation heatmap for numeric attributes and target.

4.10 Data Visualization and Observations

- The post-cleaning target distribution is imbalanced: 406 liver-disease records compared with 164 no-disease records.
- The cleaned dataset contains 430 male records and 140 female records.
- The age histogram shows patient ages distributed across a broad range.
- The box plots indicate large variation and potential extreme values in several enzyme-related measurements.
- The correlation heatmap provides a visual summary of linear relationships among numeric attributes; correlation alone does not establish causation.

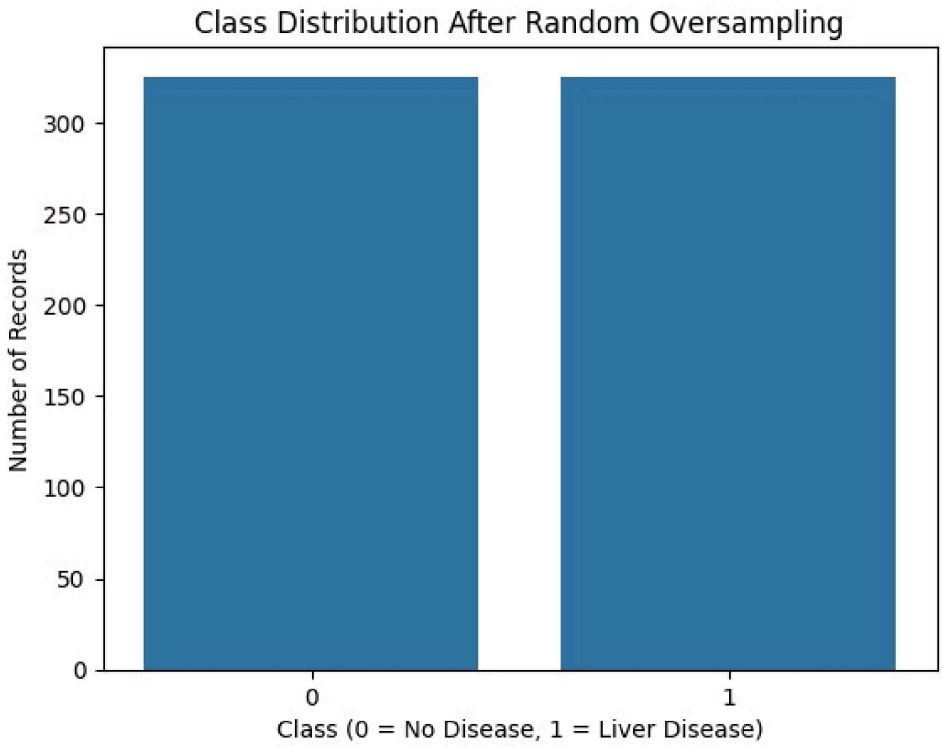


Figure 4.4: Training target distribution after RandomOverSampler.

4.11 Preprocessing Summary

Step	Method used	Result
Duplicate handling	drop_duplicates()	13 removed; 570 remain
Gender encoding	Male=1, Female=0	Numeric gender feature
Target encoding	1→1, 2→0	Positive class=1
Train-test split	Stratified 80:20	456 train / 114 test
Missing values	Median imputation	0 missing in train and test
Class balancing	RandomOverSampler on training data	325 records per class; 650 total

CHAPTER 5: PROPOSED METHODOLOGY AND ML MODEL DEVELOPMENT

5.1 Overall Project Methodology

The project follows a supervised classification workflow. The original CSV is loaded and inspected, duplicate rows are removed, categorical values are encoded, the target is separated from the predictors, and a stratified train-test split is created. Missing feature values are imputed using the training median. Random oversampling is applied only to the training data. Three classifiers are fitted and evaluated on the same untouched test subset.

5.2 Machine Learning Workflow

1. Dataset collection and loading (ILPD CSV).
2. Data inspection: shape, data types, summary statistics, missing values and duplicates.
3. Data cleaning and preparation: duplicate removal, encoding and train-test split.
4. Training-only median imputation and RandomOverSampler balancing.
5. Model training: Decision Tree, Random Forest and Gaussian Naive Bayes.
6. Prediction on the 114-record test set.
7. Evaluation using classification metrics and confusion matrices.
8. Comparison of models and selection based on F1-score and related metrics.

5.3 Selection of Machine Learning Algorithms

The three algorithms were selected because they represent different classification approaches: a single interpretable tree, an ensemble of trees and a probabilistic classifier. Comparing them helps demonstrate how different learning assumptions affect results on the same tabular dataset.

5.4 Description of Selected Algorithms

5.4.1 Decision Tree Classifier

A Decision Tree repeatedly divides the training records using conditions on input features. Each internal node represents a split, each branch represents an outcome of the split and each leaf assigns a class. It is useful as an interpretable baseline because its decision process can be visualized. In this notebook, random_state is set to 42; other tree parameters use the library defaults.

5.4.2 Random Forest Classifier

Random Forest is an ensemble method that trains multiple decision trees and combines their predictions. Combining trees can improve stability compared with relying on one tree. The notebook configures n_estimators=100 and random_state=42.

5.4.3 Gaussian Naive Bayes Classifier

Gaussian Naive Bayes applies Bayes' theorem and assumes that features are conditionally independent given the class. For continuous input attributes, it models feature values using Gaussian distributions. The notebook uses GaussianNB() with its default configuration.

5.5 Model 1 – Implementation and Training

The DecisionTreeClassifier is created with random_state=42 and fitted using X_train_balanced and y_train_balanced. Predictions are generated for X_test and stored as dt_predictions. The notebook also visualizes the trained tree to a maximum display depth of three levels.

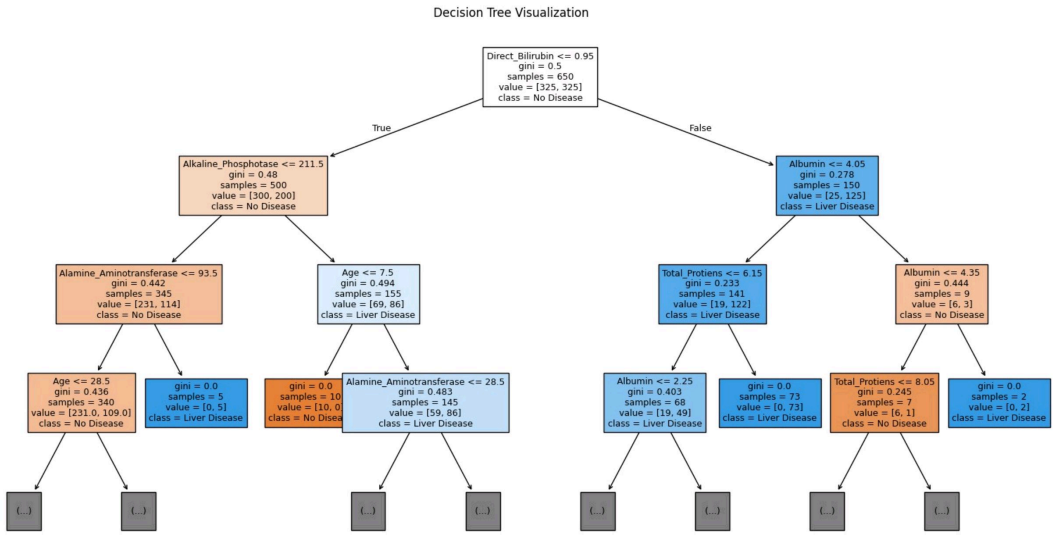


Figure 5.1: Decision Tree visualization (display limited to depth 3).

5.6 Model 2 – Implementation and Training

The RandomForestClassifier is created with n_estimators=100 and random_state=42. It is fitted on the balanced training subset and predicts the labels for the same X_test records. The resulting predictions are stored as rf_predictions.

5.7 Model 3 – Implementation and Training

The GaussianNB classifier is initialized using its default settings, fitted on the balanced training subset and used to predict the same test records. Its predictions are stored as nb_predictions.

5.8 Hyperparameter Configuration

Model	Notebook configuration
Decision Tree	DecisionTreeClassifier(random_state=42); remaining parameters default
Random Forest	RandomForestClassifier(n_estimators=100, random_state=42)
Gaussian Naive Bayes	GaussianNB(); default configuration
Data split	test_size=0.20, random_state=42, stratify=y
Imputation	SimpleImputer(strategy='median')

Oversampling	RandomOverSampler(random_state=42)

No grid search, randomized search, cross-validation-based tuning or other systematic hyperparameter optimization is recorded in the notebook.

5.9 Tools and Libraries Used

Tool / Library	Purpose
Google Colab / Jupyter Notebook	Notebook execution and project development
Python	Programming language
Pandas	CSV loading, data manipulation and tabular summaries
NumPy	Numerical operations
Matplotlib	Plotting and visualization
Seaborn	Statistical plots and heatmaps
Scikit-learn	Train-test split, imputation, classifiers and evaluation metrics
imbalanced-learn	RandomOverSampler for training-class balancing

CHAPTER 6: MODEL EVALUATION AND PERFORMANCE ANALYSIS

6.1 Evaluation Strategy

All three models are evaluated on the same 114-record test subset. The test data is not oversampled. Predictions are compared with the actual test labels using accuracy, precision, recall and F1-score. Confusion matrices are also generated with class order [0, 1], where 0 denotes No Disease and 1 denotes Liver Disease.

6.2 Evaluation Metrics

- Accuracy: proportion of all test predictions that are correct.
- Precision: proportion of predicted positive cases that are truly positive.
- Recall: proportion of actual positive cases correctly identified by the model.
- F1-score: harmonic mean of precision and recall; it balances the two measures.
- Confusion matrix: counts true negatives, false positives, false negatives and true positives.

For this project, the positive class is Liver Disease (label 1). Recall is important because it measures the fraction of liver-disease-labelled records identified by the classifier. Precision is also important because it measures how many records predicted as positive are actually positive in the test data.

6.3 Model 1 – Performance Evaluation

Decision Tree achieved accuracy 0.684, precision 0.759, recall 0.815 and F1-score 0.786 for the positive class. Its confusion matrix is [[12, 21], [15, 66]] when rows and columns are ordered as No Disease, Liver Disease.

6.4 Model 2 – Performance Evaluation

Random Forest achieved accuracy 0.711, precision 0.767, recall 0.852 and F1-score 0.807. Its confusion matrix is [[12, 21], [12, 69]]. Among the three models, it has the highest accuracy, recall and F1-score.

6.5 Model 3 – Performance Evaluation

Gaussian Naive Bayes achieved accuracy 0.570, precision 1.000, recall 0.395 and F1-score 0.566 for the positive class. Its confusion matrix is [[33, 0], [49, 32]]. It did not produce false positives in this test set, but it missed 49 of the 81 positive records, resulting in low recall.

6.6 Confusion Matrix / Actual vs. Predicted Analysis

Model	TN	FP	FN	TP
Decision Tree	12	21	15	66
Random Forest	12	21	12	69
Gaussian Naive Bayes	33	0	49	32

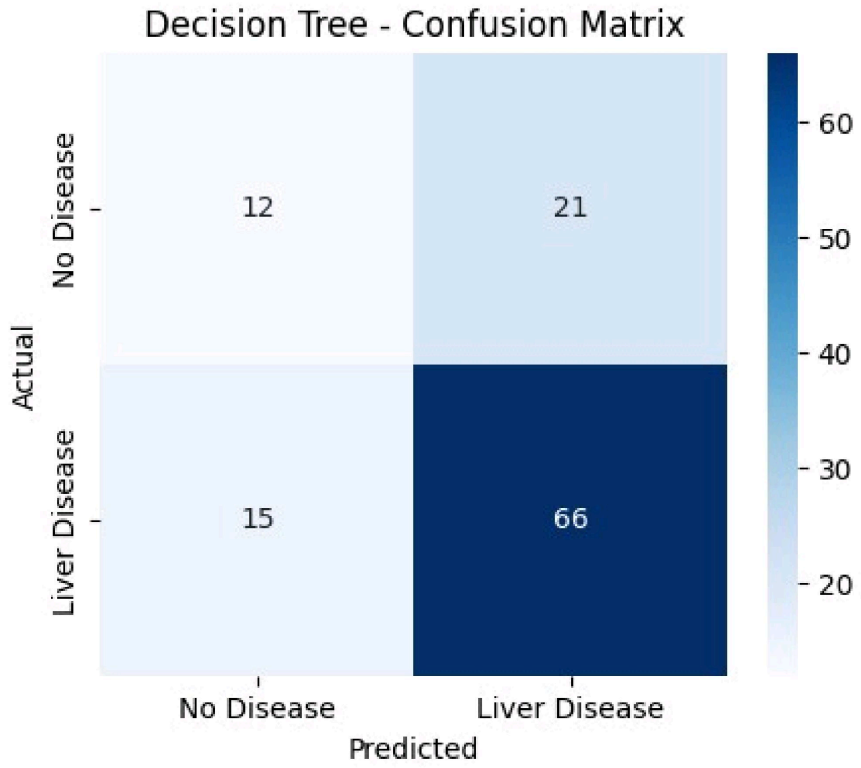


Figure 6.1: Decision Tree confusion matrix.

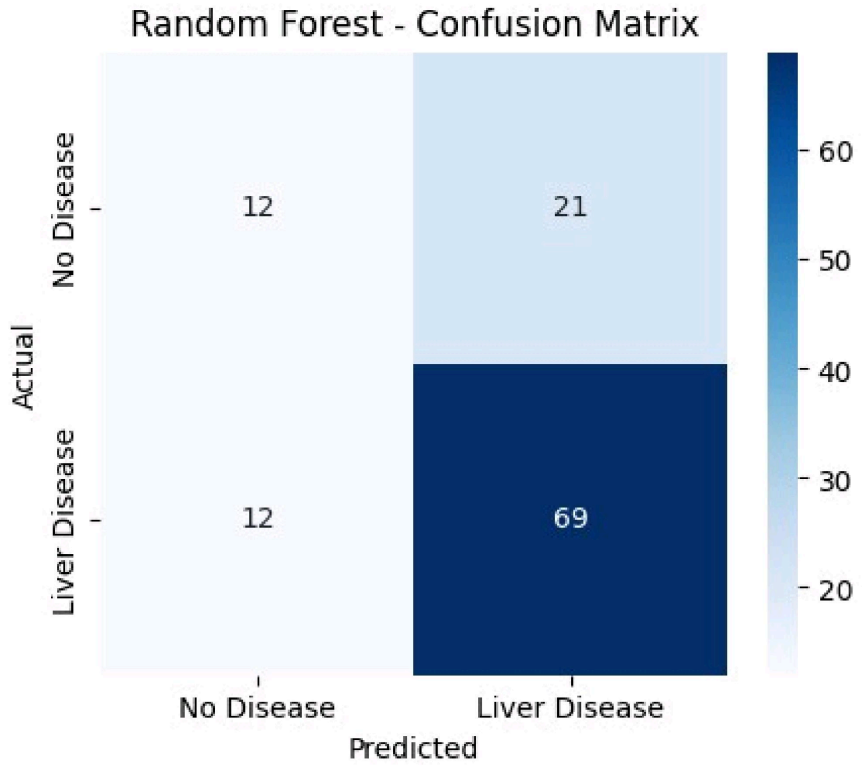


Figure 6.2: Random Forest confusion matrix.

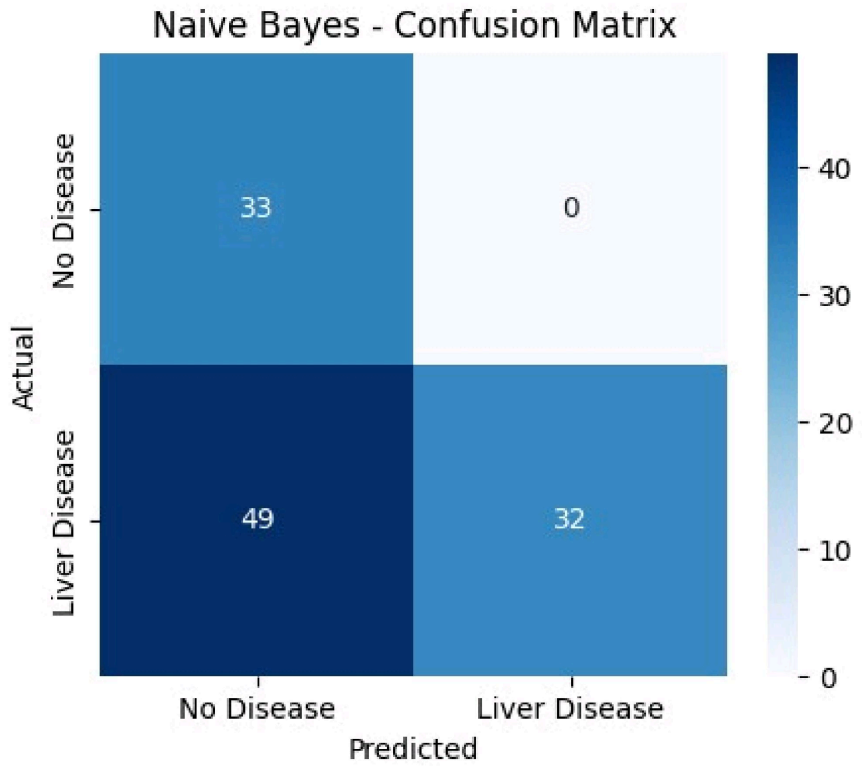


Figure 6.3: Gaussian Naive Bayes confusion matrix.

6.7 Performance Visualization

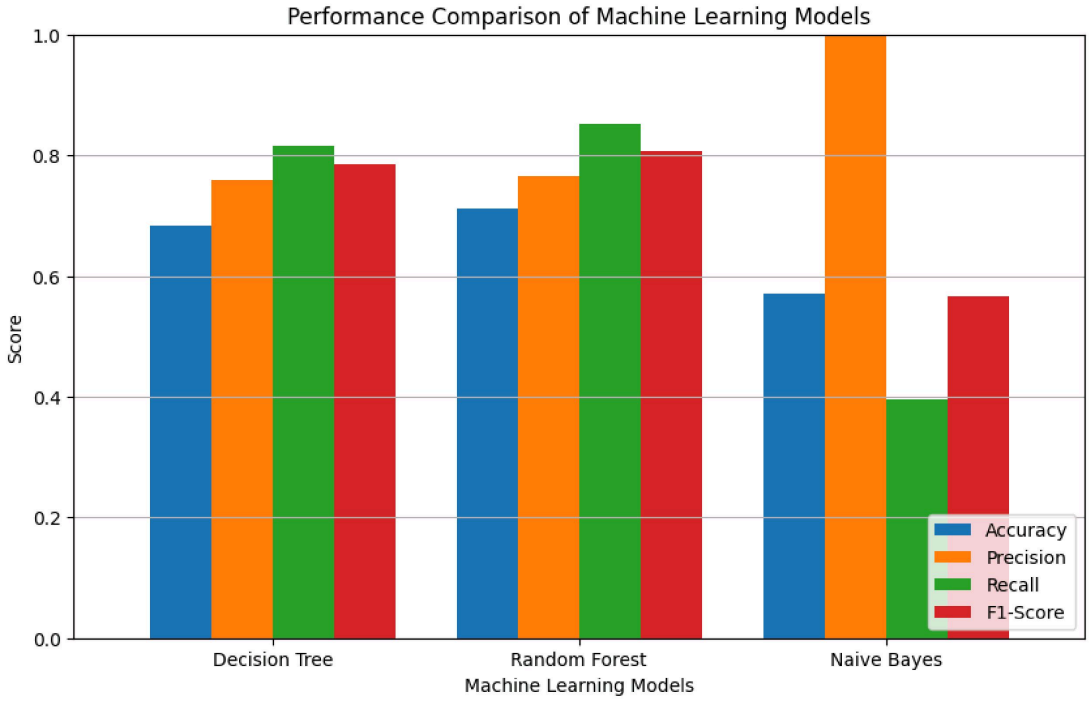


Figure 6.4: Comparison of accuracy, precision, recall and F1-score.

6.8 Comparative Analysis of Models

Metric	Decision Tree	Random Forest	Naive Bayes
Accuracy	0.6842	0.7105	0.5702
Precision	0.7586	0.7667	1.0000
Recall	0.8148	0.8519	0.3951
F1-score	0.7857	0.8070	0.5664

Random Forest provides the strongest combined precision-recall result, as indicated by the highest F1-score (0.8070). It also has the highest accuracy (0.7105) and recall (0.8519). Gaussian Naive Bayes has the highest precision (1.0000), but its recall is substantially lower. The best model therefore depends on the evaluation priority; based on the project's overall comparison and F1-score criterion, Random Forest is selected.

CHAPTER 7: RESULTS AND DISCUSSION

7.1 Experimental Results

The experiment successfully trained three classification models and generated predictions for 114 held-out test records. Random Forest produced the best overall result according to accuracy, recall and F1-score. Decision Tree produced results close to Random Forest, while Gaussian Naive Bayes showed a different precision-recall trade-off.

7.2 Model Performance Comparison

The comparative values are calculated directly in the notebook using scikit-learn metrics. The model comparison table is also displayed in the notebook and exported as model_comparison_results.csv. The values are experimental results from this particular split and should not be interpreted as guaranteed performance on new populations.

7.3 Best Performing Model

Random Forest is selected as the best-performing model among the three implemented classifiers because it achieved the highest test accuracy (0.7105) and F1-score (0.8070), along with the highest positive-class recall (0.8519). Its 69 true-positive predictions out of 81 actual positive test records were higher than the other models.

7.4 Model Optimization

Random oversampling was used to address the class imbalance in the training subset. It increased the training class counts from 325 positive and 131 negative records to 325 records in each class. No additional model hyperparameter optimization was performed. The Random Forest used 100 estimators, while the remaining model configurations were default apart from the stated random seeds.

7.5 Interpretation of Results

The Random Forest result indicates that the ensemble approach performed better than the single Decision Tree and Gaussian Naive Bayes on this test split. However, its accuracy of approximately 71% is moderate and it still makes both false-positive and false-negative errors. Gaussian Naive Bayes' precision of 1.00 should not be interpreted as overall superiority because its recall is only 0.395; it identifies fewer than half of the actual positive test records.

7.6 Key Findings

- The original file contained 583 rows and 11 columns; 13 duplicate rows were removed.

- Four missing values were present in Albumin_and_Globulin_Ratio and were imputed using the training-set median.
- The cleaned target distribution was imbalanced (406 positive versus 164 negative records).
- Random oversampling balanced the training set to 325 records in each class.
- Random Forest achieved the best F1-score (0.8070) and accuracy (0.7105).
- Naive Bayes achieved perfect positive-class precision on this test split but low recall (0.3951).

7.7 Real-World Applicability

The project demonstrates a data-science workflow that can be used for educational analysis of structured clinical records. It is not suitable for real-world diagnosis or treatment decisions. The dataset is limited in size, the evaluation uses a single train-test split, and no external clinical validation has been conducted. A clinical system would require substantially larger and more representative datasets, expert review, rigorous validation, fairness assessment, calibration and compliance with applicable medical and data-protection requirements.

7.8 Sample Record Prediction Demonstration

The notebook creates one illustrative sample record with age 45, gender encoded as 1 and selected biochemical values. The three trained models do not agree on the output:

Model	Predicted class	Model output
Decision Tree	1	Liver Disease Detected
Random Forest	1	Liver Disease Detected
Gaussian Naive Bayes	0	No Liver Disease Detected

The probability outputs for this one sample are: Decision Tree – 100.00% liver-disease class probability; Random Forest – 70.00%; Gaussian Naive Bayes – 0.04%. These are model-specific outputs for an illustrative row, not calibrated medical probabilities or a diagnosis. Disagreement between models reinforces the need for validation.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Project Summary

The project implemented a supervised machine-learning workflow to classify records in the Indian Liver Patient Dataset. It included dataset inspection, duplicate removal, categorical encoding, train-test splitting, median imputation, training-set oversampling, model development and performance comparison.

8.2 Achievement of Project Objectives

- The dataset was loaded and its structure and quality were examined.
- Missing values and duplicate rows were identified; duplicates were removed and missing feature values were imputed.
- Class imbalance was addressed in the training data using RandomOverSampler.
- Three classifiers were trained and compared using consistent test data.
- Performance was evaluated with accuracy, precision, recall, F1-score and confusion matrices.
- Random Forest was selected based on the highest F1-score and accuracy among the implemented models.

8.3 Final Conclusion

Among the three models implemented, Random Forest provided the strongest overall test-set performance, with 0.7105 accuracy, 0.7667 precision, 0.8519 recall and 0.8070 F1-score for the positive liver-disease class. Decision Tree was the second-best model by F1-score, while Gaussian Naive Bayes had high precision but comparatively low recall. The project met its objective of implementing and comparing three classification algorithms on a public benchmark dataset.

8.4 Project Limitations

- The dataset is small and may not represent all patient populations or clinical settings.
- The evaluation is based on one stratified 80:20 train-test split rather than repeated cross-validation.
- No independent external dataset was used for validation.
- No systematic hyperparameter search was performed.
- Outliers were visualized but not treated through a documented method.
- No feature scaling, feature selection or subgroup fairness analysis was implemented.
- The sample prediction is illustrative and not clinically validated.

8.5 Challenges Faced

The main implementation challenges addressed in the notebook were handling duplicate records, treating missing A/G ratio values, managing an imbalanced target distribution and comparing classifiers with different precision-recall behaviour. The use of a stratified split, training-only imputation and training-only oversampling helped organize the workflow and avoid applying resampling to the test set.

8.6 Future Enhancements

- Evaluate the models using stratified k-fold cross-validation and repeated experiments.
- Perform systematic hyperparameter tuning for each classifier.
- Test on a separate external dataset and assess generalization.
- Conduct subgroup performance analysis, including gender-stratified metrics.
- Investigate feature importance and model explanation methods.
- Evaluate probability calibration and decision thresholds.

REFERENCES / BIBLIOGRAPHY

[1] B. Ramana and N. Venkateswarlu, “ILPD (Indian Liver Patient Dataset),” UCI Machine Learning Repository, 2022. DOI: 10.24432/C5D02C. https://archive.ics.uci.edu/dataset/225/ilpd+indian+liver+patient+dataset

[2] I. Straw and H. Wu, “Investigating for bias in healthcare algorithms: a sex-stratified analysis of supervised machine learning models in liver disease prediction,” BMJ Health & Care Informatics, 2022. https://pmc.ncbi.nlm.nih.gov/articles/PMC9039354/

[3] Scikit-learn Developers, “scikit-learn: Machine Learning in Python – User Guide.” https://scikit-learn.org/stable/

[4] The pandas development team, “pandas documentation.” https://pandas.pydata.org/docs/

[5] imbalanced-learn Developers, “imbalanced-learn documentation.” https://imbalanced-learn.org/stable/

[6] Matplotlib Developers, “Matplotlib documentation.” https://matplotlib.org/stable/

[7] Seaborn Developers, “Seaborn documentation.” https://seaborn.pydata.org/

[8] Project implementation source: Liver_Disease_Prediction_PBL.ipynb, executed Google Colab supplied for this report.

APPENDIX A – DATASET INFORMATION

Dataset file used: indian_liver_patient.csv. The notebook reports 583 original rows, 11 columns, ten predictor attributes and one target column. The exact Kaggle URL from which the team downloaded the CSV was not recorded in the notebook. The official UCI dataset record and DOI are listed in Reference .

Target encoding in notebook	Meaning
1	Liver Disease
0	No Liver Disease

APPENDIX B – ADDITIONAL RESULTS

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	0.6842	0.7586	0.8148	0.7857
Random Forest	0.7105	0.7667	0.8519	0.8070
Gaussian Naive Bayes	0.5702	1.0000	0.3951	0.5664

Confusion matrix order is [[TN, FP], [FN, TP]] with class order [No Disease, Liver Disease].

Model	Confusion matrix
Decision Tree	[[12, 21], [15, 66]]
Random Forest	[[12, 21], [12, 69]]
Gaussian Naive Bayes	[[33, 0], [49, 32]]

APPENDIX C – PROJECT SCREENSHOTS

The following figures are extracted from the executed notebook outputs and included in the relevant chapters: class distribution, gender distribution, age histogram, numerical-feature box plots, correlation heatmap, oversampled training distribution, Decision Tree visualization, three confusion matrices and model-performance comparison.

APPENDIX D – GITHUB REPOSITORY

GitHub repository URL: https://github.com/Mahesh204505/liver_disease_prediction

APPENDIX E – TEAM CONTRIBUTIONS

Team member	Enrollment number	Contribution
R.Mahesh	BETN1AI25204	Data Preprocessing&Analysis,Presentation,Report
R.Barath	BETN1AI25205	Classification Models
P.Vamsi	BETN1AI25200	Model evaluation& Comparision